

Assignment - 02

Applying Microservices and Event-Driven Architecture in E-Commerce.

Scenario

Imagine developing an e-commerce platform with the following architecture.

→ Microservices Architecture:

- Service: User Management, Product catalog, Shopping cart, ~~and~~ order processing, payment, and notification.

→ Benefits: Each Service can be developed, deployed and scaled independently, enhancing flexibility and resilience.

Applying SOLID Principles

→ Single Responsibility Principle:

Each microservice handles a specific business function.

→ Open/Closed principle: Interchangeable service implementation without affecting the system's behaviour.

→ Interface Segregation principle: Services expose only necessary interfaces, preventing client from depending on unused functionalities.

→ Dependency Inversion principle: High-level modules depend on abstractions, allowing for flexible and testable code.

Observing DRY and KISS Principles

→ DRY (Don't Repeat Yourself):

- utilize shared libraries for common functionalities like logging and authentication
- centralize configuration management to avoid redundancy.

KISS (keep it Simple, stupid):

- Design Services with clear and concise responsibilities
- Avoid over-engineering: Implement features as needed.